

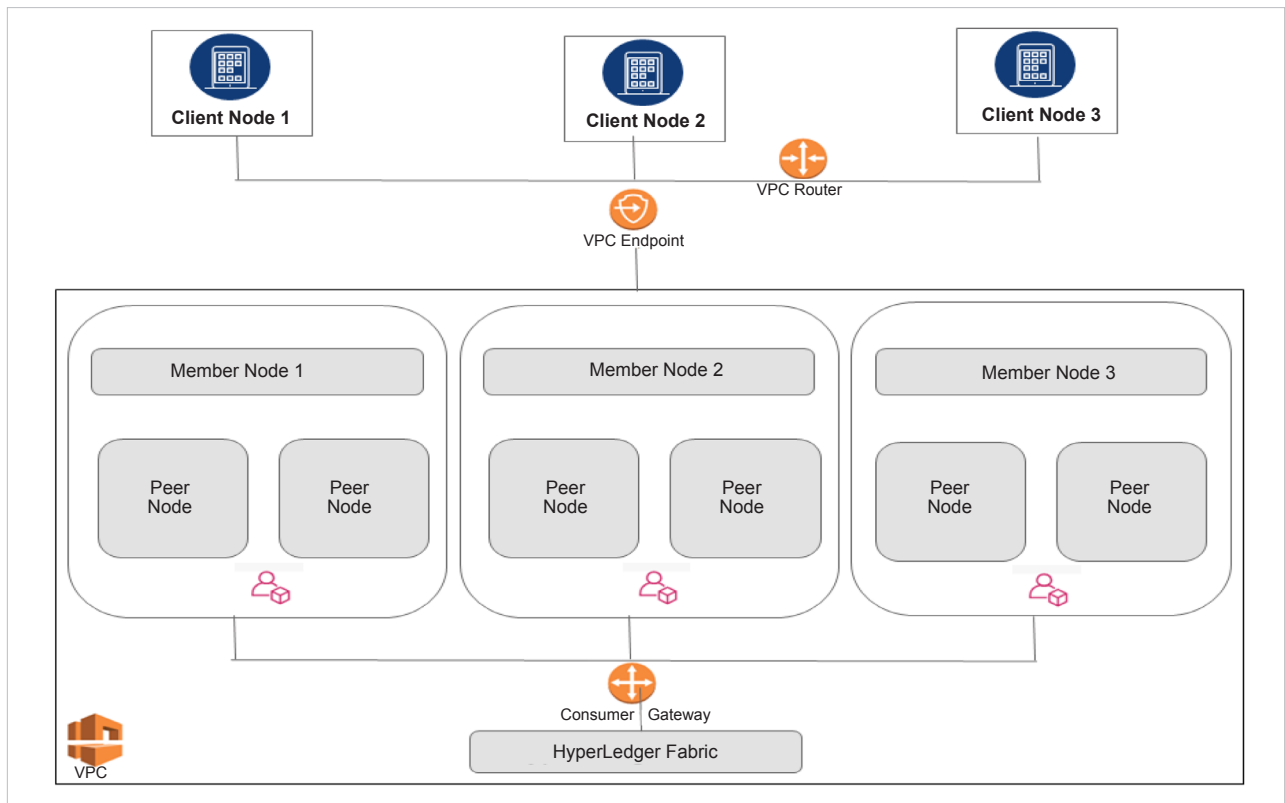


Figure 4: Blockchain Hyperledger architecture in AWS

Handling security and compliance issues in blockchain smart city administration

Smart city administration has many use cases that require an efficient platform like cloud based blockchain architecture to carry out self-healing and auto-scaling services on-the-fly in real-time.

Blockchain based smart city administration uses the advantages of a permissioned ledger based blockchain platform that are listed below.

Shared ledger: This helps to build a multi-transaction system with ‘a single point of failure’.

Consensus: This makes the transaction commit/complete only when all the parties (nodes) agree to network a verified transaction.

Immutability: This enables smart administration activity to be transparent and foolproof from tampering.

Smart contract: This sets a defined, rule based transaction communication

using a piece of code so that transactions can be coupled with validation, proof verification and tax calculation.

Provenance: Due to the nature of a distributed blockchain, the entire transaction is copied in all the nodes in the platform, and every node travel for a transaction is captured in the history

of the transaction for quickly tracking the stages.

These benefits make blockchain based smart city administrations have agility and flexibility with respect to the architecture. Using digital IDs also makes smart city transactions secure. **END** 🐧

References

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